of 1-cyanooctane occurs in about 100% yield after 1 or 2 hours. In this reaction, the tetraalkylammonium cation, which has hydrophobic and hydrophilic ends, transports CN^- from the aqueous phase to the organic phase using its hydrophilic end and facilitates the reaction with 1-chlorooctane as shown below:

$$NaCN + R_4N^+Cl^- \rightarrow R_4N^+CN^- + NaCl$$

aqueous phase — It moves to organic phas e

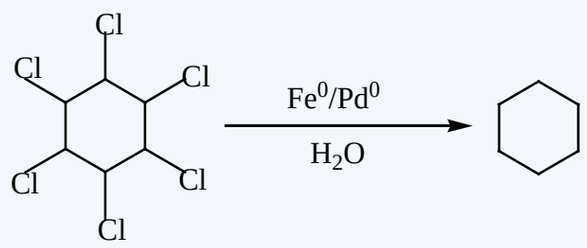
$$R_4N^+CN^- + R\text{-}Cl \rightarrow R\text{-}CN + R_4N^+Cl^-$$

Both in organic phase — organic phase — It moves to aqueous phase, releases Cl^- again pick s up CN^- and transports it.

So phase transfer catalyst, speeds up the reaction by transporting one reactant from one phase to another.

Nano Catalysis:

Nano materials such a metallic nano particles, metal oxides, etc., are used as catalyst in many chemical transformation, Nanocatalysts carry the advantages of both homogeneous and heterogeneous catalyses. Like homogeneous catalysts, the nanocatalysts give 100% selective transformations and excellent yield and show extremely high activity. Like the heterogeneous catalysts, nanocatalysts can be recovered and recycled. Nanocatalysts are actually soluble heterogeneous catalysts. An example for nanoparticles catalysed reaction is given below

(i) 

Lindane — cyclohexane

Fe^0/Pd^0 : Nanobimetallic catalyst (Zerovalent state)

10.5 Colloid, Dispersion phase and dispersion medium

Origin of study of colloid starts with Thomas Graham who observed diffusion of that a solution of sugar, urea or sodium chloride through a membrane but not glue, gelatine or gum. He called the former substances as crystalloids and the latter as colloids (In Greek, kola as gum, eidos-like).

Later it was realised that any substance can be converted into a colloid by reducing its particle size to 1-200nm.

Hence, colloid is a homogeneous mixture of two substances in which one substance (smaller proportion) is dispersed in another substance(large proportion).

In a colloid, the substance present in larger amount is called dispersing medium and the substance present in less amount is called dispersed phase.

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